Conditional compact-manifold capture for the original-clock IGATH-2 recursion

Abstract

We prove a conditional capture theorem for a fixed-parameter IGATH-2 recursion near a compact normally nondegenerate manifold of local minima. The manifold and objective are both of class C^5, and the normal Hessian spectrum lies in a fixed compact subinterval of $(0, 4)$. An independent history pair may enter at any sufficiently late original index, with uniformly small scaled normal position and current-base normal history and with any prescribed finite scaled tangent-history bound. The resulting suffix remains in a controlled tube, has finite quadratic slow-clock dissipation, and approaches the critical set of the squared norm restricted to the manifold. A discrete two-history contraction retains the entry transient, a strict discrete energy closes the capture bootstrap, and a curvature-dependent frozen target is derived and tracked to order $k^{-3/2}$. This is a newly formulated C^5/C^5 theorem. No convergence to one point, finite total travel, arbitrary early capture, or globally least-norm selection is asserted.

Keywords: IGATH-2; Tikhonov regularization; minimum manifold; conditional capture; normal stability; slow-clock dissipation

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Statement, literal map, and scope

Let M be a compact embedded C^5 submanifold without boundary of $\mathbb{R}^d$, and let F be C^5 on an open neighborhood of M. Assume, for every $p \in M$,

$$\nabla F(p) = 0, \qquad \ker \nabla^2 F(p) = T_p M, \qquad \operatorname{spec}(H_p) \subset [\mu, L], \quad 0 < \mu \le L < 4, \tag{1}$$

where $H_p = \nabla^2 F(p)|_{N_p M}$ is the self-adjoint normal restriction. Thus M is a normally nondegenerate manifold of local minima. Different connected components need not have the same critical value. Write P_p, N_p for the orthogonal tangent and normal projectors, respectively. In a sufficiently small tubular neighborhood U, let π denote the nearest-point projection. Norms are Euclidean unless another norm is specified. Set

$$q(p) = \tfrac{1}{2}|p|^2, \qquad g(p) = \operatorname{grad}_M q(p) = P_p p.$$

Our second fundamental form convention is $\mathrm{II}_p(a, b) = N_p D_a b$. Equivalently, the normal component of the acceleration of a geodesic with initial velocity a is $\mathrm{II}_p(a, a)$.

For integers $k \ge K$, start from the independent history x_{K-1}, x_K and use the literal recursion

$$\begin{aligned} y_k &= x_k + \left(1 - \frac{3}{2\sqrt{k}}\right)(x_k - x_{k-1}) - \frac{1}{4}\big(\nabla F(x_k) - \nabla F(x_{k-1})\big), \\ x_{k+1} &= y_k - \frac{1}{4}\left(\nabla F(y_k) + \frac{y_k}{k}\right). \end{aligned} \tag{2}$$

The clock is k, not $k-K$ and not a restarted index. No condition on an earlier prefix is asserted or needed. For $x_k \in U$, define

$$\begin{aligned} p_k &= \pi(x_k), & n_k &= x_k - p_k, \\ u_k &= N_{p_k}(x_k - x_{k-1}), & v_k &= \sqrt{k}\, P_{p_k}(x_k - x_{k-1}). \end{aligned} \tag{3}$$

The normal history u_k is projected at the *current* base p_k; it is not $n_k - n_{k-1}$.

The map is the IGATH-2 scheme in the final published version of Attouch, Balhag, Chbani, and Riahi [1], Section 4.2, with

$$s = \tfrac{1}{4}, \qquad \varepsilon_k = \tfrac{1}{k}, \qquad \delta = 3, \qquad \beta = \tfrac{1}{2}.$$

Indeed $\delta\sqrt{s\varepsilon_k} = 3/(2\sqrt{k})$ and $\beta\sqrt{s} = 1/4$. The gradient difference is unregularized, and the Tikhonov term acts at y_k. That source introduces the explicit scheme but leaves its convergence analysis for further study. Its continuous and proximal results are not used as a local nonconvex convergence theorem here.

Theorem 1 (Conditional original-clock capture)**.** *Under* (1)*, there exist a tubular radius $r > 0$ and a fixed $\theta > 0$, depending only on the displayed geometry and F, with the following property. For every prescribed finite $B_0 \geq 0$, there is an integer $K_0(B_0)$ such that, at every original index $K \geq K_0(B_0)$, every independent history pair satisfying*

$$x_K \in U_r, \qquad \sqrt{K}\,|n_K| \leq \theta, \qquad \sqrt{K}\,|u_K| \leq \theta, \qquad |v_K| \leq B_0 \tag{4}$$

has a well-defined suffix under (2) *for all $k \geq K$. The predecessor x_{K-1} automatically lies in the neighborhood where F is defined once K_0 is chosen; this is a consequence of* (4)*.*

The following conclusions hold.

1. *All x_k and y_k remain in a fixed controlled tubular neighborhood of M, all p_k are defined, and $|v_k| \leq B(B_0)$ for some finite $B(B_0)$.*

2. *For constants $C(B_0) < \infty$ and $0 < \rho < 1$,*

$$|n_k| + |u_k| \leq C(B_0)\theta K^{-1/2}\rho^{k-K} + \frac{C(B_0)}{k}. \tag{5}$$

3. *The slow-clock dissipation satisfies*

$$\sum_{k=K}^{\infty} k^{-1/2}\big(|v_k|^2 + |g(p_k)|^2\big) < \infty, \qquad v_k \longrightarrow 0, \quad g(p_k) \longrightarrow 0. \tag{6}$$

4. *With $\operatorname{Crit}_M q = \{p \in M : g(p) = 0\}$,*

$$\operatorname{dist}(x_k, \operatorname{Crit}_M q) \longrightarrow 0.$$

The nonempty compact omega-limit set of (x_k) is contained in $\operatorname{Crit}_M q$. The predictor sequence (y_k) has exactly the same omega-limit set.

5. *The actual normal position and normal history satisfy*

$$\begin{aligned} n_k &= k^{-1}c(p_k, v_k) + O(k^{-3/2}), \\ u_k &= -\frac{1}{2k}\mathrm{II}_{p_k}(v_k, v_k) + O(k^{-3/2}), \end{aligned} \tag{7}$$

where

$$c(p, v) = -H_p^{-1}N_p p - (4H_p^{-1} - I)\mathrm{II}_p(v, v). \tag{8}$$

In particular,

$$kn_k + H_{p_k}^{-1}N_{p_k}p_k \longrightarrow 0.$$

These are derived tracking estimates, not an assumed exact invariant branch.

The big-O constants may depend on the admitted history and its original entry index. The tube, θ, and the choice of K_0 do not depend on the admitted history pair beyond B_0 and the entry bounds.

The hypotheses and entry test of Theorem 1 are specified afresh. Its C^5/C^5 regularity is deliberately stronger than a previously suggested C^4/C^3 scope; no equivalence to an unavailable historical theorem or proof is claimed. The proof below establishes precisely the stated conditional critical-set capture.

1 Conventions and uniform geometry

Put $h_k = k^{-1/2}$, $\gamma = 3/2$, and $b = 1/4$. On N_pM, set

$$A_p = I - bH_p.$$

Its eigenvalues belong to the compact interval $[1 - L/4, 1 - \mu/4] \subset (0, 1)$.

Choose a closed tubular neighborhood of M contained in the domain of F, and a slightly larger such neighborhood still contained in that domain. Shrink their radii as needed below. Compactness and the stated regularity give uniform bounds for the required derivatives of the nearest-point projection, tangent and normal projectors, F, and the intrinsic exponential map on a uniform small geodesic neighborhood of the diagonal. All Taylor remainders below are uniform in $p \in M$. A compact C^5 embedded manifold suffices for these finite-order estimates; neither analyticity nor a Łojasiewicz inequality is used.

We use tangent and normal parallel transport along the unique short minimizing geodesic between sufficiently close points of M. Transport is isometric in each bundle. Bars over vectors based at the next point indicate transport back to the current point. Norms on pairs are Euclidean product norms. Constants $C, C_{R,D}$ and so on are finite and may increase between displays. All constants are chosen before the entry index K.

For $p \in M$ and small $\eta \in N_pM$, Taylor expansion gives

$$\nabla F(p + \eta) = H_p\eta + O(|\eta|^2). \tag{1.1}$$

Consequently, if $z = r + \eta$ with $r = \pi(z)$, then

$$|P_p \nabla F(z)| \le C\big(|r - p|\,|\eta| + |\eta|^2\big). \tag{1.2}$$

Indeed $P_r H_r \eta = 0$ and $|P_p - P_r| \le C|p - r|$. For small $\xi \in T_pM$, the geodesic expansion is

$$\exp_p \xi = p + \xi + \tfrac{1}{2}\mathrm{II}_p(\xi, \xi) + O(|\xi|^3). \tag{1.3}$$

For tangent transport τ from p to r and an ambient vector $a + c$, where $a \in T_pM$ and $c \in N_pM$, one has

$$\tau^{-1} P_r(a + c) = a + O\big(\mathrm{dist}_M(p, r)|c| + \mathrm{dist}_M(p, r)^2|a|\big). \tag{1.4}$$

To see the absence of a first-order error on a, the transported vector and its orthogonal projection onto T_rM have the same first-order ambient normal term $\mathrm{II}_p(\log_p r, a)$. Their difference is second order. The term from c is bounded by projector variation. Taylor expansion in a finite collection of local orthonormal bundle frames makes the remainder uniform.

Finally, differentiating $\nabla F(\exp_p(tv)) = 0$ twice at $t = 0$ gives

$$D^3F(p)[v, v] = -H_p \mathrm{II}_p(v, v), \qquad v \in T_pM. \tag{1.5}$$

Here $D^3F(p)[v, v]$ is the vector obtained by contracting two slots of the third derivative. This is a vector identity, not merely its normal projection.

2 A finite-history one-step estimate

We work at one original integer k. Let $h = k^{-1/2}$, $p = \pi(x)$, $n = x - p$, and decompose the actual history $d = x - x_{\mathrm{prev}}$ as

$$d = hv + u, \qquad v \in T_pM, \quad u \in N_pM.$$

Fix finite bounds R, D, and assume $|v| \le R$ and $|n| + |u| \le Dh$. If h is sufficiently small depending on R, D, the predecessor, predictor, and next point all belong to the larger chosen tube. Their projections are defined, and $p_+ = \pi(x_+)$ is in the uniform geodesic neighborhood of p. Write

$$z = (n, u), \qquad z_+ = (n_+, u_+), \qquad v_+ = h_+^{-1} P_{p_+}(x_+ - x), \quad h_+ = (k + 1)^{-1/2}.$$

The following estimates are uniform:

$$\bar{z}_+ = B_p z + R^N, \qquad |R^N| \le C_{R,D} h^2, \qquad B_p = \begin{pmatrix} A_p & A_p^2 \\ A_p - I & A_p^2 \end{pmatrix}, \tag{2.1}$$

$$\log_p p_+ = hv + r^p, \qquad |r^p| \le C_{R,D} h^2, \tag{2.2}$$

$$\bar{v}_+ = v - h\big(\gamma v + bg(p)\big) + r^v, \qquad |r^v| \le C_{R,D}\left(h^2 + |z| + \frac{|z|^2}{h}\right). \tag{2.3}$$

The bars in (2.1) and (2.3) mean normal and tangent transport, respectively.

Proof of the one-step estimates. The point $x_{\text{prev}} = p+n-u-hv$ is $O(h)$ from p. The projection Taylor formula at p bounds its normal displacement from M by $C(|z|+h^2)$, and its projected base is $O(h)$ from p. The analogous assertions for y, x_+ follow initially from their $O(h)$ proximity to p; the refined normal bounds follow from their defining formulas and (1.1). More explicitly, first-order gradient expansion at p gives

$$\nabla F(x) - \nabla F(x_{\text{prev}}) = H_p u + O(h^2), \qquad y - p = hv + n + A_p u + O(h^2).$$

The normal displacement of y from M is $n + A_p u + O(h^2)$, so

$$x_+ - p = hv + A_p n + A_p^2 u + O(h^2).$$

Subtracting the $O(h^2)$ normal component of $p_+ - p$ gives

$$\bar{n}_+ = A_p n + A_p^2 u + O(h^2).$$

Moreover,

$$u_+ = N_{p_+}\big[(p_+ - p) + (n_+ - n)\big].$$

The first term is $O(h^2)$, and transport and projector changes on n are $O(h|n|)$. Hence

$$\bar{u}_+ = (A_p - I)n + A_p^2 u + O(h^2),$$

which proves (2.1).

The tangential estimate requires more than an undifferentiated $O(h^2)$ gradient error. By (1.2), the normal displacement bounds just obtained, and the $O(h)$ base changes,

$$|P_p \nabla F(x)| \le C|n|^2,$$
$$|P_p \nabla F(x_{\text{prev}})| + |P_p \nabla F(y)| \le C_{R,D}\big(h|z| + |z|^2 + h^3\big).$$

Also $P_p y = g(p) + hv + O(h^2)$; here $g(p)$ is the tangent projection of the ambient position p itself. Applying P_p to the actual increment therefore yields

$$\begin{aligned} P_p(x_+ - x) &= hv - \gamma h^2 v - bh^2 g(p) + E, \\ |E| &\le C_{R,D}\big(h^3 + h|z| + |z|^2\big). \end{aligned} \tag{2.4}$$

This uses the unregularized gradient difference and the Tikhonov term at y exactly as in (2).

The normal component $N_p(x_+ - x)$ is $O(|z| + h^2)$. By (1.4), replacing the current-base tangent projection of the actual increment by its back-transported next-base tangent projection adds at most $C(h|z| + h^3)$. Division by h_+ and

$$h/h_+ = \sqrt{1+h^2} = 1 + O(h^2)$$

prove (2.3).

For position, project $x_+ - x = (p_+ - p) + (n_+ - n)$ tangentially at p. The projection of n is zero; that of n_+ has size at most $Ch(|z| + h^2)$. Formula (1.3) says that $P_p(p_+ - p)$ differs from $\log_p p_+$ by $O(h^3)$. Equation (2.4) and $|z| \le Dh$ now prove (2.2). No continuous-time limit has replaced the actual recursion. ∎

3 Contraction of the normal two-history system

For a scalar eigenvalue $a \in (0,1)$ of A_p, the corresponding block of B_p is

$$B(a) = \begin{pmatrix} a & a^2 \\ a-1 & a^2 \end{pmatrix}.$$

Its characteristic polynomial is $r^2 - a(1+a)r + a^2$. The discriminant is $a^2((1+a)^2 - 4) < 0$, and the two roots have modulus $a < 1$. Since a ranges over a fixed compact subinterval of $(0,1)$, the matrices B_p are uniformly exponentially stable. In detail, the distinct conjugate roots are uniformly separated on that interval, so complex diagonalization gives

$$|B_p^j| \le C\sigma^j \qquad (p \in M,\ j \ge 0)$$

for one $\sigma < 1$. Orthogonal diagonalization of H_p reduces the full matrix to these blocks without changing the norm.

On the normal pair bundle define

$$Q_p = \sum_{j=0}^{\infty} (B_p^j)^* B_p^j.$$

The series converges uniformly, as do the finitely many derivatives needed here: derivatives of powers introduce polynomial factors in j multiplying a uniform geometric bound. Thus Q has sufficient regularity, and

$$I \le Q_p \le M_Q I, \qquad B_p^* Q_p B_p = Q_p - I.$$

There is $\rho_* < 1$ with $|B_p z|_{Q_p} \le \rho_* |z|_{Q_p}$. Since $p_+ - p = O(h)$, the change of Q under normal transport is $O(h)$. Fix ρ with $\rho_* < \rho < 1$. For small enough h, (2.1) gives

$$|z_+|_{Q_{p_+}} \le \rho |z|_{Q_p} + C_{R,D} h^2. \tag{3.1}$$

This contracts the two-history phase; an estimate on n alone would not suffice.

Iterating while the cap assumptions hold, with $j = k - K$, gives

$$|z_k| \le C\theta h_K \rho^j + C_{R,D} h_k^2. \tag{3.2}$$

Indeed, the initial Q-norm is at most $\sqrt{2M_Q}\theta h_K$, and

$$\sum_{i=K}^{k-1} \frac{\rho^{k-1-i}}{i} \le \frac{1}{k} \sum_{\ell=0}^{\infty} \rho^\ell (\ell + 2) \le \frac{C_\rho}{k}.$$

Here $k/i = 1 + (k-i)/i \le \ell + 2$ for $\ell = k - 1 - i$ and $i \ge 1$. Constants cover $k = K$ as well. In particular,

$$\sup_{j\ge 0} \frac{h_K}{h_{K+j}} \rho^j \le \sup_{j\ge 0} \sqrt{1+j}\, \rho^j < \infty.$$

Choose D larger than four times the resulting coefficient of θ in the bound for $(|n|+|u|)/h$, plus 4. Here θ may be any one fixed number in $(0, 1]$, chosen once for the theorem. Increasing K later makes the forcing contribution $C_{R,D}h_k$ smaller than 1 uniformly. Thus (3.2) improves the cap $|n| + |u| \le Dh$ to a strict interior cap, for example $(D/2)h$.

Equations (2.3), (3.2), and $|z| \le Dh$ imply

$$|r_k^p| \le Ch_k^2, \qquad |r_k^v| \le Ch_k^2 + t_k, \qquad t_k = C\theta h_K \rho^{k-K}. \tag{3.3}$$

These constants may depend on the prescribed tangential cap, but not on the admitted history. In particular, $\sum_{k\ge K} t_k \le C\theta h_K$. The entry transient remains explicit; the normal variables have not been assumed to be $O(1/k)$ immediately upon entry.

4 A strict slow-clock Lyapunov function

Let

$$G = \max_M |g|, \qquad Q_{\max} = \max_M q, \qquad D_q = \max_M |\operatorname{Hess}_M q|.$$

Choose a fixed $\epsilon > 0$ with

$$\epsilon \le 1, \qquad \epsilon(D_q + 9/2) \le 3/4.$$

On TM, define

$$V(p, v) = \tfrac{1}{2}|v|^2 + bq(p) + \epsilon\langle g(p), v\rangle.$$

Use the intrinsic vector field

$$p' = v, \qquad \nabla_t v = -\gamma v - bg(p)$$

only as an auxiliary differentiation device. Its exact derivative of V is

$$DV = -\gamma|v|^2 + \epsilon \operatorname{Hess}_M q[v, v] - \epsilon\gamma\langle g, v\rangle - \epsilon b|g|^2.$$

Since

$$\gamma|g|\,|v| \le \frac{b}{2}|g|^2 + \frac{\gamma^2}{2b}|v|^2, \qquad \frac{\gamma^2}{2b} = \frac{9}{2},$$

we obtain

$$DV \le -\tfrac{3}{4}|v|^2 - \tfrac{\epsilon}{8}|g|^2 \le -c\big(|v|^2 + |g|^2\big), \qquad c = \min(3/4, \epsilon/8) > 0. \tag{4.1}$$

The cross term supplies dissipation of the restricted-norm gradient as well as velocity. No isolated-critical-point or convexity assumption on $q|_M$ is needed.

We now prove an inequality for the actual finite-history iterates. On every fixed velocity cap, the horizontal and vertical first derivatives satisfy

$$|D_p V| + |D_v V| \le C(|v| + |g|),$$

and the second derivatives are uniformly bounded. Write $A_0 = |v| + |g|$. Apply second-order Taylor expansion in the exponential and parallel-transport coordinates of (2.2)–(2.3). The leading increment is $h(v, -\gamma v - bg)$, with residual increments r^p, r^v. Equations (3.3) and (4.1) give

$$\begin{aligned} V(p_+, v_+) - V(p, v) \le & -ch(|v|^2 + |g|^2) + Ch^2 A_0^2 + Ch^2 A_0 \\ & + Ct_k + C(h^4 + t_k^2). \end{aligned}$$

Terms multiplying t_k are uniformly bounded on the fixed cap; for large K, $t_k \le 1$ and $t_k^2 \le t_k$. For small h, half of the negative term absorbs the first positive term. Young's inequality gives

$$Ch^2 A_0 \le \frac{c}{4} h(|v|^2 + |g|^2) + C'h^3.$$

Also $h^4 \le h^3$. Decreasing the dissipation constant if needed, we conclude that

$$\begin{aligned} V(p_{k+1}, v_{k+1}) - V(p_k, v_k) \le & -c_1 h_k \big(|v_k|^2 + |g(p_k)|^2\big) \\ & + Ch_k^3 + Ct_k, \end{aligned} \tag{4.2}$$

for a fixed $c_1 > 0$. Both positive error sequences are summable:

$$\sum_{k=K}^{\infty} h_k^3 \le Ch_K, \qquad \sum_{k=K}^{\infty} t_k \le C\theta h_K.$$

This is an absorbed, summable discrete energy estimate. No bare $O(h^2)$ additive energy error is being summed.

5 Closing the capture bootstrap

Fix the prescribed B_0. The entry energy satisfies

$$V(p_K, v_K) \le V_0 := \tfrac{1}{2} B_0^2 + bQ_{\max} + \epsilon G B_0.$$

Since $q \ge 0$, completion of a square gives, for every p, v,

$$V(p, v) \ge \tfrac{1}{4}|v|^2 - \epsilon^2 G^2. \tag{5.1}$$

Choose finite $R > B_0 + 2$ so large that

$$\tfrac{1}{4}(R-1)^2 - \epsilon^2 G^2 > V_0 + 3.$$

Choose D as in Section 3. Put

$$\beta = \sup_{p \in M} \|B_p\|, \qquad D_{\text{aux}} = \max\{D+1, 2\beta D + 2\},$$

where the operator norm is the Euclidean product norm. This local notation β is independent of the source algorithm's parameter notation above. All one-step, contraction, Taylor, and energy constants are now fixed on the working caps R, D and, when necessary, the larger finite caps $R+1, D_{\text{aux}}$. Choose the original integer $K_0(B_0)$ sufficiently large that:

1. all one-step constructions are valid on these caps;
2. all points within the estimated $O(h)$ distances of M stay in the chosen domain and tubes;
3. the strict normal-cap improvement in Section 3 holds;
4. the summed positive errors in (4.2) from any $K \geq K_0$ are less than 1;
5. the raw next-step bounds below place the next velocity in the $R+1$ cap and the next scaled normal state in the D_{aux} cap.

Here are input-only justifications of the last condition. Section 2 constructs the predecessor, predictor, next point, and next projection in the larger tube before any next-state phase cap is assumed. Isometry of normal transport and (2.1) give

$$|z_+|_2 \leq \beta |z|_2 + C_{R,D} h^2.$$

The sum norm is at most $\sqrt{2}$ times the Euclidean product norm, and $h/h_+ \leq \sqrt{2}$ for $k \geq 1$. Thus

$$\frac{|n_+| + |u_+|}{h_+} \leq 2\beta D + 2C_{R,D} h < D_{\text{aux}}$$

for large K_0. Likewise, (2.3), $|z| \leq Dh$, and isometry of tangent transport imply

$$\big||v_+| - |v|\big| \leq C_{R,D} h < 1,$$

so $|v_+| < R+1$. Neither raw estimate uses a strict next-state working cap. The requirements are finitely many smallness conditions on h_K with constants fixed before K, so such an integer K_0 exists. This choice preserves the original clock.

The entry test gives $|v_K| \leq B_0 < R - 1$ and

$$|n_K| + |u_K| \leq 2\theta h_K < D h_K.$$

It also gives

$$|x_{K-1} - p_K| \leq |n_K| + |u_K| + h_K |v_K| \leq (2\theta + B_0) h_K,$$

placing the predecessor in the domain without any additional hypothesis.

Induct up to a possible first violation of the tube or working phase caps. Suppose the working caps hold through index k. The input-only one-step construction and raw estimates first define the next point and place it in the larger tube and auxiliary phase caps. Apply (3.1) to this just-constructed step and iterate through index $k+1$. This requires only the current working caps and the $O(h_k)$ displacement of the already-defined next base; it does not assume the next normal state is in the working cap. Equation (3.2) therefore improves that next state to

$$|n_{k+1}| + |u_{k+1}| \le (D/2)h_{k+1}.$$

The raw bound $|v_{k+1}| < R+1$ supplies the tangent-bundle neighborhood for the energy Taylor estimate on the same step. Its errors use the current normal bound and historical transient, not a next-state working-cap assumption. Consequently (4.2) holds through this step. Summing through index k gives

$$V(p_{k+1}, v_{k+1}) \le V_0 + 1.$$

By (5.1) and the choice of R, this improves the tangent bound to $|v_{k+1}| < R-1$. The normal distance is at most $(D/2)h_{k+1}$, and the predictor normal distance on the constructed step is at most $C(Dh_k + h_k^2)$, so the tube bounds improve as well for large K_0. A first violation is impossible. This proves all-time existence and control of the actual suffix, together with (3.2).

Since V is bounded below, summing (4.2) gives

$$\sum_{k=K}^{\infty} h_k\big(|v_k|^2 + |g(p_k)|^2\big) < \infty. \tag{5.2}$$

The argument works for each prescribed finite B_0. Increasing B_0 changes the phase cap and how late the original entry index must be. It does not restrict p_K to a particular component, well, or basin on M.

6 Vanishing phase and critical-set capture

The phase recurrences imply

$$\big||v_{k+1}| - |v_k|\big| \le Ch_k + t_k,$$

and tangent transport τ_k from p_k to p_{k+1} gives

$$|g(p_{k+1}) - \tau_k g(p_k)| \le C\,\mathrm{dist}_M(p_{k+1}, p_k) \le Ch_k.$$

The ratio estimate of Section 3 bounds $t_k \le Ch_k$ uniformly for $k \ge K$. Hence, for $a_k = |v_k|^2 + |g(p_k)|^2$,

$$|a_{k+1} - a_k| \le Ch_k. \tag{6.1}$$

Together with $a_k \geq 0$, $\sum h_k a_k < \infty$, $h_k \to 0$, and $\sum h_k = \infty$, this implies $a_k \to 0$.

For a direct discrete proof, suppose $a_k \geq \delta > 0$ at arbitrarily large indices. Equation (6.1) keeps $a_j \geq \delta/2$ on the ensuing slow-time interval until $\sum h_j$ first reaches $\delta/(2C)$, apart from an endpoint error that vanishes as $k \to \infty$. Each such interval contributes a uniformly positive amount to $\sum h_j a_j$. Infinitely many disjoint intervals may be selected, unless a stays bounded below forever; the latter also contradicts summability because $\sum h_j = \infty$. Both possibilities are impossible.

Thus $v_k \to 0$ and $g(p_k) \to 0$. Compactness makes $\mathrm{Crit}_M\, q$ nonempty and compact. It also implies

$$\mathrm{dist}(p_k, \mathrm{Crit}_M\, q) \longrightarrow 0 :$$

otherwise a subsequence staying a positive distance away would have a convergent subsequence with zero gradient and positive distance from the critical set. Since $n_k \to 0$, the same distance conclusion holds for x_k.

The suffix stays in a bounded tube, so its omega-limit set is nonempty and compact. All cluster points belong to $\mathrm{Crit}_M\, q$. Finally,

$$d_k = h_k v_k + u_k \longrightarrow 0.$$

The gradients at x_k and x_{k-1} differ by $O(|d_k|)$ in the controlled neighborhood. Equation (2) therefore gives $y_k - x_k \to 0$. Bounded sequences whose pointwise difference tends to zero have identical omega-limit sets, proving the predictor assertion.

7 Curvature correction and normal tracking

Equation (3.2) already implies $|z_k| = O(h_k^2)$ as $k \to \infty$, because the exponential transient is eventually dominated by every power of k. All estimates in this section are uniform on the controlled velocity cap; their constants may depend on the admitted entry.

At a sufficiently late step write $h = h_k$, $p = p_k$, $v = v_k$, $A = A_p$, $H = H_p$, and $S = \mathrm{II}_p(v, v)$. Since $n, u = O(h^2)$, Taylor expansion at p and (1.5) give

$$\nabla F(x) - \nabla F(x_{\mathrm{prev}}) = Hu + \tfrac{1}{2} h^2 HS + O(h^3). \tag{7.1}$$

Explicitly, $x_{\mathrm{prev}} = p + n - u - hv$, so

$$\begin{aligned} \nabla F(x_{\mathrm{prev}}) &= H(n-u) + \tfrac{1}{2} h^2 D^3 F(p)[v, v] + O(h^3) \\ &= H(n-u) - \tfrac{1}{2} h^2 HS + O(h^3), \end{aligned}$$

whereas $\nabla F(x) = Hn + O(h^4)$. This verifies the curvature sign in (7.1).

The predictor has tangent coordinate $hv - \gamma h^2 v + O(h^3)$ at p and ambient normal coordinate

$$n + Au - \tfrac{b}{2} h^2 HS + O(h^3).$$

Its normal displacement relative to its own projected base, represented back at p to the needed order, is therefore

$$n + Au - \tfrac{1}{2}h^2(I + bH)S + O(h^3). \tag{7.2}$$

The normal part of $x_+ - p$ at the old base equals

$$An + A^2u + \tfrac{b^2}{2}h^2H^2S - bh^2N_pp + O(h^3).$$

By (1.3) and $\log_p p_+ = hv + O(h^2)$, the normal part of $p_+ - p$ is $h^2S/2 + O(h^3)$. Subtracting yields

$$\bar{n}_+ = An + A^2u - \tfrac{1}{2}h^2A(I + bH)S - bh^2N_pp + O(h^3), \tag{7.3}$$

using $I - b^2H^2 = A(I + bH)$.

The normal projection at the *new* base of the chord $p_+ - p$ is $-h^2S/2 + O(h^3)$, represented at p. This sign differs from its old-base normal projection. Thus

$$\bar{u}_+ = \bar{n}_+ - n - \tfrac{1}{2}h^2S + O(h^3). \tag{7.4}$$

Transport errors here are $O(h^3)$ because the normal vectors have size $O(h^2)$ and the base moves by $O(h)$. These formulas retain curvature from both the gradient-difference term and the new-base history convention.

Equations (7.3)–(7.4) take the form

$$\bar{z}_+ = B_pz + h^2f(p, v) + O(h^3),$$

where

$$f_n = -\tfrac{1}{2}A(I + bH)S - bN_pp, \qquad f_u = f_n - \tfrac{1}{2}S.$$

Solve the algebraic frozen equation $\zeta = B_p\zeta + f$. Its unique solution is

$$\begin{aligned}
\zeta_u &= -\tfrac{1}{2}S, \\
\zeta_n &= -H^{-1}N_pp - b^{-1}H^{-1}AS \\
&= -H^{-1}N_pp - (4H^{-1} - I)\mathrm{II}_p(v, v).
\end{aligned}$$

For verification, substitute $\zeta_u = -S/2$ into the first row. Its curvature coefficient becomes

$$-\tfrac{1}{2}\big[A^2 + A(I + bH)\big]S = -AS,$$

because $A + I + bH = 2I$. Hence

$$(I - A)\zeta_n = -AS - bN_pp,$$

which gives the displayed formula since $I - A = bH$. The second row reduces to $\zeta_u = -S/2$. Invertibility also follows from the strict stability of B_p.

The algebraic target is not assumed invariant. It has sufficient regularity on the bounded tangent-bundle region. The established phase estimates and $|z| = O(h^2)$ give

$$\operatorname{dist}_M(p_+, p) = O(h), \qquad \bar{v}_+ - v = O(h), \qquad h_+^2 - h^2 = O(h^4).$$

Consequently,

$$\overline{h_+^2 \zeta(p_+, v_+)} - h^2 \zeta(p, v) = O(h^3). \tag{7.5}$$

Set $e_k = z_k - h_k^2 \zeta(p_k, v_k)$. Equations (7.3)–(7.5), the Q-norm construction, and a possibly later fixed index J imply

$$|e_{k+1}|_{Q_{p_{k+1}}} \le \rho |e_k|_{Q_{p_k}} + C h_k^3, \qquad k \ge J,$$

for a fixed $\rho < 1$, enlarged from its earlier value if necessary. Iteration gives $|e_k| = O(h_k^3)$: the initial exponential term is eventually $O(h_k^3)$, and the forcing convolution is of this order because

$$(k/i)^{3/2} \le (k - i + 1)^{3/2}$$

and a geometric sequence multiplied by this polynomial is summable. This proves both expansions in (7).

Finally $v_k \to 0$, and H_p^{-1} and II_p are uniformly bounded, so

$$\begin{aligned} k n_k + H_{p_k}^{-1} N_{p_k} p_k &= -(4 H_{p_k}^{-1} - I) \mathrm{II}_{p_k}(v_k, v_k) + O(k^{-1/2}) \\ &\longrightarrow 0. \end{aligned}$$

The curvature-dependent target is needed in the intermediate bounded-velocity regime, although its curvature contribution vanishes in the eventual critical-capture limit. This completes the proof of Theorem 1.

8 Scope and exclusions

Entry is conditional on a checkable independent history at a sufficiently late *original* index; capture from arbitrary early data is not asserted. The C^5/C^5 assumptions replace the suggested C^4/C^3 scope. No isolated-critical-point or analytic hypothesis is assumed. Capture is of $\operatorname{Crit}_M q$; one-point convergence, finite path length, and global least-norm selection do not follow from the quadratic slow-clock dissipation. No unavailable historical definitions, constants, or proofs are claimed as recovered.

References

[1] H. Attouch, A. Balhag, Z. Chbani, and H. Riahi. Accelerated gradient methods combining Tikhonov regularization with geometric damping driven by the Hessian. *Applied Mathematics & Optimization* **88**, article 29 (2023). doi:10.1007/s00245-023-09997-x. Final published version, Section 4.2, boxed IGATH-2 scheme.